Proof. Since f is continuous on the compact set $\mathcal{B}_B$, it is uniformly continuous. Choose $\eta > 0$ such that $\|z - z'\|_\infty \le \eta$ implies $|f(z) - f(z')| \le \delta/2$.

Partition each coordinate interval $[-B, B]$ into subintervals of length at most η and form the induced rectangular grid on $\mathcal{B}_B$. Triangulate each grid cell into finitely many simplices (e.g., by ordering coordinates). Define $\tilde{f}$ by setting $\tilde{f}$ to equal f at every grid vertex and extending $\tilde{f}$ linearly over each simplex.

Then $\tilde{f}$ is continuous and piecewise-linear. For any $z \in \mathcal{B}_B$, let v be a vertex of the simplex containing z. By construction, $\|z - v\|_\infty \le \eta$, hence

$$|f(z) - f(v)| \le \delta/2.$$

Moreover, $\tilde{f}(v) = f(v)$ for all vertices v. Since $\tilde{f}$ is an affine combination of its vertex values on each simplex, there exists some vertex v of that simplex such that

$$|\tilde{f}(z) - \tilde{f}(v)| \le \max_{v' \text{ vertex of simplex}} |\tilde{f}(v') - \tilde{f}(v)| = \max_{v' \text{ vertex}} |f(v') - f(v)|.$$

All vertices of a simplex lie within ℓ_∞-distance at most η from each other, so by uniform continuity $|f(v') - f(v)| \le \delta/2$. Therefore $|\tilde{f}(z) - f(v)| \le \delta/2$ and

$$|f(z) - \tilde{f}(z)| \le |f(z) - f(v)| + |f(v) - \tilde{f}(z)| \le \delta/2 + \delta/2 = \delta.$$

□

D.3. ReLU representation of piecewise-linear functions

Next we use that two-layer ReLU networks represent continuous piecewise-linear functions.

Lemma D.2 (Two-layer ReLU realizes PL functions)**.** *Let $\tilde{f} : \mathbb{R}^D \to \mathbb{R}$ be continuous and piecewise-linear with finitely many linear regions. Then there exist $m \in \mathbb{N}$, $w_1, \dots, w_m \in \mathbb{R}^D$, $a_1, \dots, a_m \in \mathbb{R}$, and an affine map $\ell(z) = \langle c, z\rangle + d$ such that*

$$\tilde{f}(z) \;=\; \ell(z) + \sum_{r=1}^{m} a_r\, \sigma(\langle w_r, z\rangle) \qquad \textit{for all } z \in \mathbb{R}^D.$$

Equivalently, $\tilde{f}$ can be written as $W_2\sigma(W_1 z + b_1) + b_2$ for suitable matrices/vectors.

Proof. A standard representation result for continuous piecewise-linear functions states that any such $\tilde{f}$ admits a "hinge" decomposition into an affine function plus a finite sum of signed ReLU ridge functions. One way to see this is to observe that the set of break hyperplanes (where the gradient changes) is finite; across each such hyperplane the function has an affine kink. Each kink can be expressed by a term of the form $a\,\sigma(\langle w, z\rangle)$ (possibly after subtracting its affine part), and summing all kinks together with a base affine function yields $\tilde{f}$ exactly.

For completeness, we give an explicit reduction to the two-layer form. Write the above representation as

$$\tilde{f}(z) = \langle c, z\rangle + d + \sum_{r=1}^{m} a_r\, \sigma(\langle w_r, z\rangle).$$

Let

$$W_1 := \begin{bmatrix} w_1^\top \\ \vdots \\ w_m^\top \end{bmatrix} \in \mathbb{R}^{m\times D}, \qquad b_1 := 0 \in \mathbb{R}^m, \qquad W_2 := \begin{bmatrix} a_1 & \cdots & a_m \end{bmatrix} \in \mathbb{R}^{1\times m},$$

and set $b_2 := d$ and incorporate the affine term $\langle c, z\rangle$ by using the identity $\langle c, z\rangle = \langle c, \sigma(z)\rangle - \langle c, \sigma(-z)\rangle$, which is itself a two-layer ReLU network (Lemma D.4 below). Combining these gives a two-layer ReLU realization of $\tilde{f}$. □

D.4. Main arithmetic approximation lemma (vector-valued)

We now combine the previous lemmas and extend coordinate-wise to $\mathbb{R}^D$ outputs.

Lemma D.3 (Token-wise ReLU FFN approximates continuous updates)**.** *Fix $B > 0$. Let $f : \mathcal{B}_B \to \mathbb{R}^D$ be continuous and write $f = (f_1, \ldots, f_D)$. For any $\delta > 0$, there exists a two-layer ReLU network $\phi : \mathbb{R}^D \to \mathbb{R}^D$ of the form $\phi(z) = W_2\sigma(W_1 z + b_1) + b_2$ such that*

$$\sup_{z \in \mathcal{B}_B} \|\phi(z) - f(z)\|_\infty \;\le\; \delta.$$

Proof. Let $\delta > 0$ be given. For each coordinate $i \in [D]$, apply Lemma D.1 to f_i with accuracy δ to obtain a continuous piecewise-linear $\tilde{f}_i$ with

$$\sup_{z \in \mathcal{B}_B} |f_i(z) - \tilde{f}_i(z)| \le \delta.$$

By Lemma D.2, each $\tilde{f}_i$ can be realized *exactly* by a two-layer ReLU network $\phi_i : \mathbb{R}^D \to \mathbb{R}$.

Define $\phi : \mathbb{R}^D \to \mathbb{R}^D$ by stacking the coordinate networks:

$$\phi(z) := \big(\phi_1(z), \ldots, \phi_D(z)\big).$$

This is again a two-layer ReLU network (block-diagonal construction of W_2), and for every $z \in \mathcal{B}_B$,

$$\|\phi(z) - f(z)\|_\infty = \max_{i \in [D]} |\phi_i(z) - f_i(z)| = \max_{i \in [D]} |\tilde{f}_i(z) - f_i(z)| \le \delta.$$

□

D.5. Useful explicit gadget: exact affine maps

We also record a simple exact representation that is occasionally convenient.

Lemma D.4 (Exact affine maps)**.** *For any matrix $U \in \mathbb{R}^{D \times D}$ and vector $c \in \mathbb{R}^D$, there exists a two-layer ReLU network $\psi(z) = \widetilde{W}_2\sigma(\widetilde{W}_1 z + \widetilde{b}_1) + \widetilde{b}_2$ such that $\psi(z) = Uz + c$ for all $z \in \mathbb{R}^D$.*

Proof. Use $z = \sigma(z) - \sigma(-z)$ coordinate-wise. Let

$$\widetilde{W}_1 = \begin{bmatrix} I \\ -I \end{bmatrix} \in \mathbb{R}^{2D \times D}, \quad \widetilde{b}_1 = 0, \quad \widetilde{W}_2 = \begin{bmatrix} U & -U \end{bmatrix} \in \mathbb{R}^{D \times 2D}, \quad \widetilde{b}_2 = c.$$

Then $\widetilde{W}_2\sigma(\widetilde{W}_1 z) = U\sigma(z) - U\sigma(-z) = Uz$, hence $\psi(z) = Uz + c$. □

D.6. How this lemma is used

In the executor, routing contributes a read vector $\mathcal{R}_\tau(q; P)$ to a designated work token via residual addition (see Appendix B). Thus the input to the FFN at that token lies in a bounded box $\mathcal{B}_B$ by design, encoding both the current work state and the freshly read parameters. Lemma D.3 then guarantees that a fixed token-wise FFN can approximate the desired local update uniformly to within δ, which is the per-step arithmetic error used in the composition/error-budget argument.

E. Composition and Error Budget Template

This appendix provides a reusable template for composing routing and token-wise computation while controlling the end-to-end error. The purpose is not to optimize constants, but to provide a clear bookkeeping scheme for allocating local error budgets across macro-steps and turning per-step guarantees into a uniform end-to-end bound.

E.1. Abstract macro-step model

Fix a prompt $p \in P$ and an input $x \in X$. Consider a designated *work token* whose state after macro-step t is denoted by $s_t \in \mathbb{R}^m$ for some fixed dimension m (typically $m \le D$). We model one macro-step as

$$s_{t+1} \;=\; \Psi_t(s_t,\ \lambda_t(p))\,, \qquad t = 0, 1, \ldots, T_{\text{macro}} - 1, \tag{5}$$

where:

- $\Psi_t : \mathbb{R}^m \times \mathbb{R}^{r_t} \to \mathbb{R}^m$ is the *ideal* update map at macro-step t.
- $\lambda_t(p) \in \mathbb{R}^{r_t}$ denotes the ideal parameter fragment for step t encoded in the prompt.

In the construction, $\lambda_t(p)$ is obtained by prompt reads and simple linear post-processing, and Ψ_t is implemented (approximately) by token-wise FFNs.

Let $\widehat{s}_t$ denote the corresponding state produced by the actual executor. We abstract a realized macro-step as

$$\widehat{s}_{t+1} \;=\; \widehat{\Psi}_t\Big(\widehat{s}_t,\; \widehat{\lambda}_t(p,x)\Big), \qquad t = 0, 1, \ldots, T_{\text{macro}} - 1, \tag{6}$$

where $\widehat{\lambda}_t(p,x)$ is the routed-and-assembled parameter fragment (possibly depending on x through the query generation), and $\widehat{\Psi}_t$ is the realized token-wise computation at step t.

E.2. Per-step error decomposition

We isolate two local error sources at step t.

(i) Routing error. We assume that the assembled fragment $\widehat{\lambda}_t(p,x)$ is close to the ideal fragment $\lambda_t(p)$:

$$\|\widehat{\lambda}_t(p,x) - \lambda_t(p)\|_\infty \;\le\; \delta_t^{\text{route}} \qquad \text{for all } (p,x) \in P \times X. \tag{7}$$

In our proofs, (7) is obtained by applying the routing bounds in Appendix C to each prompt read and union-bounding over at most R reads per macro-step.

(ii) Arithmetic (computation) error. We assume that, given an *ideal* fragment, the realized computation approximates the ideal update uniformly:

$$\sup_{s \in \mathcal{S}} \big\|\widehat{\Psi}_t(s, \lambda_t(p)) - \Psi_t(s, \lambda_t(p))\big\|_\infty \;\le\; \delta_t^{\text{arith}}, \tag{8}$$

where $\mathcal{S} \subset \mathbb{R}^m$ is a bounded state set containing all intermediate states (typically $\mathcal{S} = [-B, B]^m$). In our proofs, (8) follows from Appendix D by approximating $\Psi_t(\cdot, \lambda_t(p))$ with a two-layer ReLU FFN to accuracy δ_t^{arith} uniformly over $\mathcal{S}$.

(iii) Sensitivity to routed parameters. We also record a mild Lipschitz-type sensitivity condition of the ideal update in its second argument: there exists $L_t \ge 0$ such that for all $s \in \mathcal{S}$ and all admissible λ, λ',

$$\|\Psi_t(s,\lambda) - \Psi_t(s,\lambda')\|_\infty \;\le\; L_t\, \|\lambda - \lambda'\|_\infty. \tag{9}$$

In the construction, Ψ_t is built from affine maps and bounded nonlinearities, so such an L_t exists.

E.3. A one-step bound

Lemma E.1 (One-step error propagation)**.** *Assume* (7)*,* (8)*, and* (9)*. In addition, assume that the ideal update Ψ_t is K_t-Lipschitz in the work-state argument on the relevant domain:*

$$\|\Psi_t(s,\lambda) - \Psi_t(s',\lambda)\|_\infty \le K_t \|s - s'\|_\infty \qquad \textit{for all } s, s' \textit{ and admissible } \lambda. \tag{10}$$

Then for any t and any $(p,x) \in P \times X$,

$$\|\widehat{s}_{t+1} - s_{t+1}\|_\infty \;\le\; K_t\, \|\widehat{s}_t - s_t\|_\infty \;+\; \delta_t^{\text{arith}} \;+\; L_t\, \delta_t^{\text{route}}.$$

In particular, if $K_t \le 1$ (as enforced in our executor design), this reduces to the additive bound $\|\widehat{s}_{t+1} - s_{t+1}\|_\infty \le \|\widehat{s}_t - s_t\|_\infty + \delta_t^{\text{arith}} + L_t \delta_t^{\text{route}}$.

Proof. Add and subtract $\Psi_t(\widehat{s}_t, \lambda_t(p))$ and $\widehat{\Psi}_t(\widehat{s}_t, \lambda_t(p))$:

$$\begin{aligned}
\|\widehat{s}_{t+1} - s_{t+1}\|_\infty &= \big\|\widehat{\Psi}_t(\widehat{s}_t, \widehat{\lambda}_t) - \Psi_t(s_t, \lambda_t)\big\|_\infty \\
&\le \big\|\widehat{\Psi}_t(\widehat{s}_t, \widehat{\lambda}_t) - \widehat{\Psi}_t(\widehat{s}_t, \lambda_t)\big\|_\infty \\
&\quad + \big\|\widehat{\Psi}_t(\widehat{s}_t, \lambda_t) - \Psi_t(\widehat{s}_t, \lambda_t)\big\|_\infty \\
&\quad + \big\|\Psi_t(\widehat{s}_t, \lambda_t) - \Psi_t(s_t, \lambda_t)\big\|_\infty.
\end{aligned}$$